FLATNESS AND FINITE PRESENTATION OF THE INERTIA, AND SURJECTIVITY OF THE DIAGONAL, FOR A GERBE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The fibre product description of the inertia of a base change. Invoked inside the counted proof (source lines 6456-6459). Allows the inertia of the base-changed stack to be computed from that of the original.
- The descent statements for flatness and for local finite presentation. Invoked inside the counted proof (source lines 6460-6462 and 6481-6484). Reduce the properties of the inertia and of the diagonal to a flat finitely presented cover.
- The characterization of a gerbe by flatness and local finite presentation of its inertia. Invoked inside the counted proof (source lines 6463-6465). Gives flatness and local finite presentation of the inertia of the base-changed stack.
- The local structure of a gerbe as a quotient stack by a flat group space. Invoked inside the counted proof (source lines 6470-6473). Realises the base-changed stack as a quotient in order to compute the diagonal.

Source reference. Original lemma lemma-base-change-gerbe: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6129–6129. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-gerbe, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-cartesian-square-inertia: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1436–1436. Navigation pointer only: the counted text refers to the Stacks item lemma-cartesian-square-inertia, which is an existing labeled result of this same chapter and is not

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Source reference. Original lemma lemma-descent-finite-presentation: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5833–5833. Navigation pointer only: the counted text refers to the Stacks item lemma-descent-finite-presentation, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-descent-flat: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5389–5389. Navigation pointer only: the counted text refers to the Stacks item lemma-descent-flat, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-local-structure-gerbe: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6248–6248. Navigation pointer only: the counted text refers to the Stacks item lemma-local-structure-gerbe, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original proposition proposition-when-gerbe: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 6319–6319. Navigation pointer only: the counted text refers to the Stacks item proposition-when-gerbe, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks which makes $\mathcal{X}$ a gerbe over $\mathcal{Y}$. Then*

(1) *$\mathcal{I}_{\mathcal{X}/\mathcal{Y}} \to \mathcal{X}$ is flat and locally of finite presentation,*
(2) *$\mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ is surjective, flat, and locally of finite presentation,*
(3) *given algebraic spaces T_i, $i = 1, 2$ and morphisms $x_i : T_i \to \mathcal{X}$, with $y_i = f \circ x_i$ the morphism*
$$T_1 \times_{x_1, \mathcal{X}, x_2} T_2 \longrightarrow T_1 \times_{y_1, \mathcal{Y}, y_2} T_2$$
is surjective, flat, and locally of finite presentation,
(4) *given an algebraic space T and morphisms $x_i : T \to \mathcal{X}$, $i = 1, 2$, with $y_i = f \circ x_i$ the morphism*
$$Isom_{\mathcal{X}}(x_1, x_2) \longrightarrow Isom_{\mathcal{Y}}(y_1, y_2)$$
is surjective, flat, and locally of finite presentation.

Proof. Proof of (1). Choose a scheme Y and a surjective smooth morphism $Y \to \mathcal{Y}$. Set $\mathcal{X}' = \mathcal{X} \times_{\mathcal{Y}} Y$. By Lemma lemma-cartesian-square-inertia we obtain cartesian squares

$$\begin{array}{ccccc} \mathcal{I}_{\mathcal{X}'} & \longrightarrow & \mathcal{X}' & \longrightarrow & Y \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{I}_{\mathcal{X}/\mathcal{Y}} & \longrightarrow & \mathcal{X} & \longrightarrow & \mathcal{Y} \end{array}$$

By Lemmas lemma-descent-flat and lemma-descent-finite-presentation it suffices to prove that $\mathcal{I}_{\mathcal{X}'} \to \mathcal{X}'$ is flat and locally of finite presentation. This follows

from Proposition proposition-when-gerbe (because $\mathcal{X}'$ is a gerbe over Y by Lemma lemma-base-change-gerbe).

Proof of (2). With notation as above, note that we may assume that $\mathcal{X}' = [Y/G]$ for some group algebraic space G flat and locally of finite presentation over Y, see Lemma lemma-local-structure-gerbe. The base change of the morphism $\Delta : \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ over $\mathcal{Y}$ by the morphism $Y \to \mathcal{Y}$ is the morphism $\Delta' : \mathcal{X}' \to \mathcal{X}' \times_Y \mathcal{X}'$. Hence it suffices to show that Δ' is surjective, flat, and locally of finite presentation (see Lemmas lemma-descent-flat and lemma-descent-finite-presentation). In other words, we have to show that

$$[Y/G] \longrightarrow [Y/G \times_Y G]$$

is surjective, flat, and locally of finite presentation. This is true because the base change by the surjective, flat, locally finitely presented morphism $Y \to [Y/G \times_Y G]$ is the morphism $G \to Y$.

Proof of (3). Observe that the diagram

$$\begin{array}{ccc} T_1 \times_{x_1, \mathcal{X}, x_2} T_2 & \longrightarrow & T_1 \times_{y_1, \mathcal{Y}, y_2} T_2 \\ \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & \mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \end{array}$$

is cartesian. Hence (3) follows from (2).

Proof of (4). This is true because

$$Isom_{\mathcal{X}}(x_1, x_2) = (T \times_{x_1, \mathcal{X}, x_2} T) \times_{T \times T, \Delta_T} T$$

hence the morphism in (4) is a base change of the morphism in (3). □